

MULTIFREQUENCY ULTRASONIC INTERFEROMETER

(FOR THE MEASUREMENT OF ULTRASONIC VELOCITY IN LIQUIDS)

1. WORKING PRINCIPLE

An Ultrasonic Interferometer is a simple and direct device to determine the ultrasonic velocity in liquids with a high degree of accuracy.

The principle used in the measurement of velocity (v) is based on the accurate determination of the wavelength (λ) in the medium. Ultrasonic waves of known frequency (f) are produced by a quartz crystal fixed at the bottom of the cell. These waves are reflected by a movable metallic plate kept parallel to the quartz crystal. If the separation between these two plates is exactly a whole multiple of the sound wavelength, standing waves are formed in the medium. This acoustic resonance gives rise to an electrical reaction on the generator driving the quartz crystal and the anode current of the generator becomes a maximum.

If the distance is now increased or decreased and the variation is exactly one half wavelength ($\lambda/2$) or multiple of it, anode current becomes maximum. From the knowledge of wavelength (λ) the velocity (v) can be obtained by the relation:

$$\text{Velocity} = \text{Wavelength} \times \text{Frequency}$$

$$v = \lambda \times f$$

2. DESCRIPTION

The Ultrasonic Interferometer consists of the following parts:

- a) The High Frequency Generator : One
- b) The Measuring Cells: 11 nos
- c) Base to hold the Cell
- d) Shielded Co-ax Cable

a) The High Frequency Generator is designed to excite the quartz crystal fixed at the bottom of the measuring cell at its resonant frequency to generate ultrasonic waves in the experimental liquid filled in the " Measuring Cell ". A micrometer to observe the changes in current and two controls for the purpose of sensitivity regulation and initial adjustment of the micrometer are provided on the panel of the High Frequency Generator.

b) The Measuring Cell is specially designed double walled cell for maintaining the temperature of the liquid constant during the experiment. A fine micrometer screw has been provided at the top, which can lower or raise the reflector plate in the liquid in the liquid in the cell through a known distance. It has a quartz crystal fixed at its bottom.

3. ADJUSTMENTS OF ULTRASONIC INTERFEROMETER

For initial adjustment two knobs are provided on high frequency generator, one is marked with "ADJ" to adjust the position of the needle on the Ammeter and the knob marked "GAIN" is used to increase the sensitivity of the instrument for greater deflection, if desired.

The ammeter is used to notice the number of maximum deflections while micrometer is moved up and down in liquid.

4. PROCEDURE

1. Unscrew the knurled cap of cell and lift it away from double walled construction of the cell. In the middle portion of it pour experimental liquid and screw the knurled cap. *Wipe out excess liquid overflowing from the cell.*
2. Insert the cell in the heavy base socket and clamp it with the help of a screw provided on its side.
3. Connect the High Frequency Generator with cell by co-axial cable provided with the instrument. *In Multifrequency Ultrasonic Interferometer Model M-84, frequency selector knob should be positioned at desired frequency (same frequency as that of liquid cell chosen).*
4. Move the micrometer slowly in either clockwise or anticlockwise direction till the anode current on the ammeter on the High Frequency Generator shows a maximum or minimum. If needle is

going out of scale, adjust the deflection within scale using “Adj” and “GAIN” knob.

5. Note the readings of micrometer corresponding to the maximum or minimum (whichever is sharper) in micro-ammeter. Take about 50 readings of consecutive maximum or minimum and tabulate them as shown below:

S. No.	Micrometer Reading corresponding to Maximum / Minimum (in mm)	Difference between consecutive maxima/minima ($\lambda/2$)
1	N1	
2	N2	N2 – N1
3	N3	N3 – N2
-		
-		
50	N50	N50 – N49

6. Take average of all the differences ($\lambda/2$)
7. Once the wavelength (λ) is known the velocity (v) in the liquid can be calculated with the help of the relation

$$v = \lambda \times f$$

5. **PRECAUTIONS:**

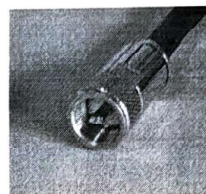
- A. Do not switch on the generator without filling the experimental liquid in the cell.

8. TROUBLE SHOOTING

A) If needle is lying before 0 (zero) perform below mentioned check points:

1. Valve: Open the top lid and check whether front tube (valve) is glowing. If valve is not glowing, switch off the main unit, remove the tube from its base and place it again. Now switch ON the unit and check whether tube is glowing. If not, please contact us for it's replacement.

2. Coaxial Cable: Check the resistance between the central pin of both the connectors with the help of the multimeter available in lab. Resistance should be less than 10 ohm. Similarly check the resistance between the outer sleeves.



3. Liquid Cell: Put the liquid cell upside down (without top reflector assembly) and check the resistance between the central hex metal portion and outer wall of the cell. It should show high resistance ≥ 5 M ohm.

4. Tuning of Generator: Assemble the setup as per the instruction manual. There are two screws on the back side of the high frequency generator. Switch ON the unit and rotate the screw A clockwise (from front direction) slightly (say around by 25 degree), observe the motion of the needle. If it comes within the scale, leave the screw position as it is and perform the experiment.

If needle doesn't come within scale, rotate screw A anticlockwise approximately same as rotated earlier and leave it as it is. Now rotate the screw B in clockwise direction, needle should move to right. If not, please contact the manufacturer.

B) If needle is lying beyond 100 (hundred) perform below mentioned check points:

1. Valve: Open the top lid and check whether back tube (valve) is glowing. If valve is now glowing, switch off the main unit, remove the tube from its base and place it again. Now switch ON the unit and check whether tube is glowing. If not, please contact us for it's replacement.

2. Tuning of Generator: Assemble the setup as per the instruction manual. There are two screws on the back side of the high frequency generator. Switch ON the unit and rotate the screw B anticlockwise (from front direction) slightly (say around by 25 degree), observe the

motion of the needle. If it comes within the scale, leave the screw position as it is and perform the experiment.

If needle doesn't come within scale, rotate screw A clockwise approximately same as rotated earlier and leave it as it is. Now rotate the screw A anticlockwise, needle should move to right. If not, please contact the manufacturer.

1.

SAMPLE CALCULATIONS

Sample : Water

$$\text{Frequency } f = 2 \text{ MHz}$$

$$\text{Average } \lambda/2 = 0.375 \text{ mm}$$

Ultrasonic velocity in sample,

$$v = \lambda \times f = 1480 \text{ m/sec}$$

$$\text{Density of liquid } \rho = 996.458 \text{ kg/m}^3$$

Compressibility β_{ad}

$$\begin{aligned} &= \frac{1}{\rho v^2} \\ &= \frac{1}{996.458 \times (1480)^2} \\ &= 4.58 \times 10^{-10} \text{ N/m}^2 \end{aligned}$$

Note 1. Above calculations are for 2 MHz Ultrasonic Interferometer. Check the frequency of Interferometer from Specifications and calculate velocity etc. as given above.

2. You may take other liquids of your choice.

3. Compressibility is temperature dependent. We may provide suitable water bath on cost to vary the temperature for calculations of compressibility at different temperatures.

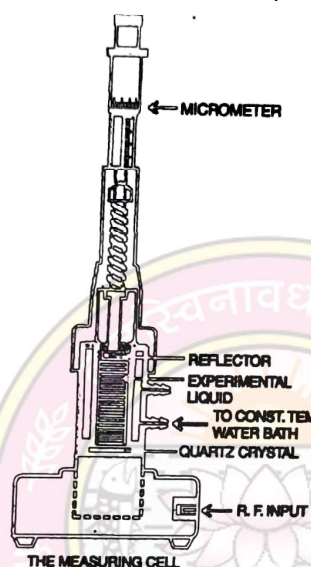


Fig 1. Cross-Section of the Liquid Cell

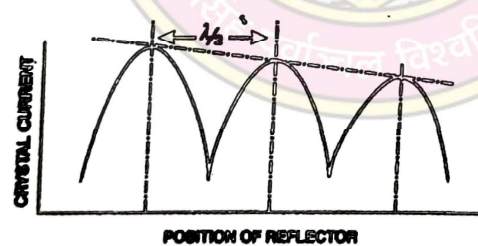


Fig 2. Position of Reflector vs Crystal Current

Note: Extra peaks in between minima and maxima may occur due to a number of reasons, but they don't affect the value of $\lambda/2$.

WORKING MANUAL FOR LABORATORY CIRCULATING WATER BATH**➤ CONSTRUCTION:-**

Double walled outer MS sheet duly powder coated. Inner S.Steel 34 grade fitted with glasswool Insulation.

➤ FITTED WITH:-

1. Lid:-Made of Stainless Steel 304 grade.
2. Circulation:-Provided through 1/20 HP for water circulation.
3. Control:-The Water Bath Temperature is controlled by Digital Temperature Controller Cum Indicator.

➤ CONTROL PANEL:-

Consist of:- 1.Digital Temperature Controller cum indicator.

2. Mains Indicator (Red).
- 3.Main Switch.
4. Switch for pump (at extreme right).

➤ INSTALLATION:-

1. First of all fill the water in the bath for necessary testing. It should be atleast 3" above the perforated SS Tray lying inside the bath. Maximum it should be 2" below the opening at the top.
2. Connect the Bath to 220 /250V, AC supply and switch it ON.
3. Press the red button (marked as set point) and set the temperature by knob(at the left of red button).
4. Green i.e. Temperature Indicator will begin to glow indicating loud ON & the unit will automatically control the Temperature of Bath.
5. Do not change the setting of T/P switch. If Temp. is not maintained at the desired temperature, keep T/P switch between 4 to 5 with the help of screw driver. After achieving temperature nearer to set point adjust T/P switch at low position (2 or 1) to avoid fluctuation in temperature.
6. Avoid any contamination in water. Preferably use double distilled water. If water is contaminated or not be used for few days, drain out the water.

➤ **TECHNICAL DETAIL:-**

1. Temperature Range:- Ambient to 85 °C
2. Accuracy > 1 °C or better.
3. Temperature Sensor:- PT 100 Thermocouple.
4. Maximum Load:- 500w.

➤ **FAULT FINDING:-**

No Power from MAINS:-

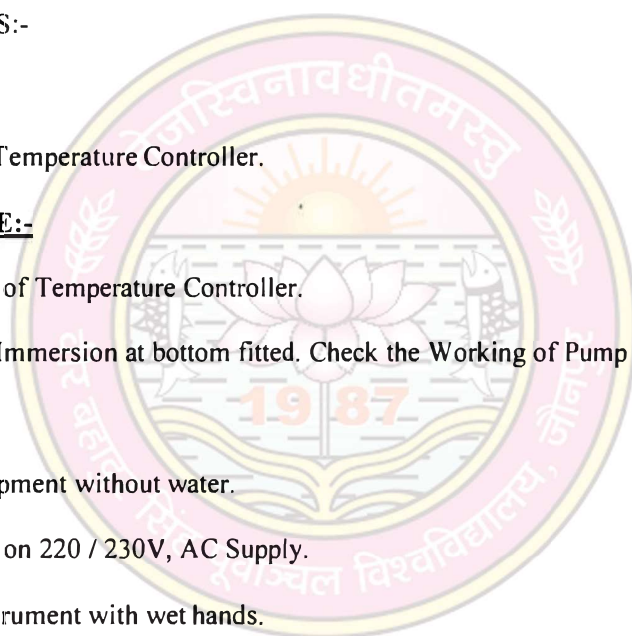
1. Check Mains Switch.
2. Check SSR / Relay of Temperature Controller.

➤ **CONTROL FAILURE:-**

1. Check the Relay / SSR of Temperature Controller.
2. Check the Working of Immersion at bottom fitted. Check the Working of Pump Stirrer.

➤ **PRECAUTION:-**

1. Never operate the Equipment without water.
2. Operate the Equipment on 220 / 230V, AC Supply.
3. Avoid touching the Instrument with wet hands.



**Actual measured Frequencies of Multifrequency Ultrasonic Interferometer
M-84(SI No 1819274)**

Frequency	Actual Measured Frequency	Tolerance
1 MHz	0.9879 MHz	± 0.0001 MHz
2 MHz	2.0070MHz	± 0.0001 MHz
3 MHz	2.9799 MHz	± 0.0001 MHz
4 MHz	4.0193 MHz	± 0.0001 MHz
5 MHz	4.9209MHz	± 0.0001 MHz
6 MHz	5.8579 MHz	± 0.0001 MHz
7 MHz	6.8687 MHz	± 0.0001 MHz
8 MHz	7.9925MHz	± 0.0001 MHz
9 MHz	8.8209 MHz	± 0.0001 MHz
10 MHz	10.1212MHz	± 0.0001 MHz
12 MHz	11.9311MHz	± 0.0001 MHz